Daily Logic Drill #1

Sheet by David Akinyele-Aje

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Statements & Quantifiers (∀/∃) · Negating Compound Statements · Negating Quantified Statements · Order of Quantifiers.

Section A Rapid Recognition

1. Negate the nested statement: "$\forall x \in \mathbb{R},\ \exists y \in \mathbb{R} : x^2 + y^2 < 1$."

Ans: $\exists x \in \mathbb{R},\ \forall y \in \mathbb{R} : x^2 + y^2 \geq 1$.

2. Over the positive integers $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$, determine the truth value of:

$$\forall n \in \mathbb{Z}^+,\ \exists m \in \mathbb{Z}^+ : m < n$$

Ans: False.

3. State the negation of the implication: "If n is even, then n is a multiple of 4 or n is a multiple of 6."

Ans: "n is even, and n is not a multiple of 4, and n is not a multiple of 6."

4. Over the real numbers $\mathbb{R}$, determine the truth value of:

$$\exists x \in \mathbb{R},\ \forall y \in \mathbb{R} : xy = 0$$

Ans: "True."

5. Over the non-zero real numbers $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$, determine the truth value of:

$$\forall y \in \mathbb{R}^*,\ \exists x \in \mathbb{R}^* : xy = 1$$

Ans: "True."

6. Given propositions P and Q, if the biconditional $P \iff Q$ is false, what is the truth value of $(P \wedge \neg Q) \vee (\neg P \wedge Q)$?

Ans: "True."

7. Negate the condition: $x \in [-2, 5) \setminus \{0\}$.

Ans: "$x < -2$ or $x = 0$ or $x \geq 5$."

8. Determine the complete set of real values of k for which the statement is true:

$$\forall x \in \mathbb{R}, \quad (x < 0 \implies x^2 > k)$$

Ans: $k \leq 0$

9. If the conditional statement $(P \wedge Q) \implies (R \vee S)$ is false, state the unique truth assignment for the propositions P, Q, R, S.

Ans: P is True, Q is True, R is False, and S is False.

10. Negate: "For every even integer $n > 2$, there exist primes p and q such that $n = p + q$."

Ans: "There exists an even integer $n > 2$ such that for all primes p and q, $n \neq p + q$."

Section B Manipulation Drills

1. An integer $n > 1$ is defined to be an $\mathcal{S}$-*number* if every factor $d > 1$ of n is a multiple of some element of $\mathcal{S} = \{2, 3\}$. Which of the following is the correct definition of an integer $n > 1$ that is *not* an $\mathcal{S}$-number? *(after TMUA Specimen Paper 2 Q17)*

Ans: B) There exists a factor $d > 1$ of n such that $\gcd(d, 6) = 1$.

2. Let P, Q, R be propositions. Suppose you are given that the compound statement

$$(P \implies Q) \wedge (Q \implies R) \wedge (R \implies P)$$

is false, but **exactly two** of the three implications are true. Can P, Q, and R all have the same truth value? *(after TMUA 2020 Paper 2 Q20)*

Ans: No.

3. Let S be the set of real numbers x such that:

$$\forall y \in (0, 1), \quad xy < x + y$$

Determine the set S in interval notation. *(after TMUA 2022 Paper 2 Q13)*

Ans: $[0, \infty)$.

4. Negate the statement: "$\forall x > 0,\ \exists y > 0 : (y < x \wedge xy \geq 1)$." Is the original statement true or false over $\mathbb{R}$?

Ans: Negation: "$\exists x > 0,\ \forall y > 0 : (y \geq x \vee xy < 1)$." Original is False.

5. Three statements about an integer n are given:

I. n is a multiple of 6.

II. n is odd.

III. n is a prime strictly greater than 3.

Given that **exactly one** of I, II, III is true, which statement is impossible to be the true one? *(after TMUA 2016 Paper 2 Q4)*

Ans: Statement III.

6. Negate the nested statement: "$\exists p$ prime such that $\forall q$ prime with $q > p,\ q$ is odd." Is the original statement true or false?

Ans: Negation: "For every prime p, there exists a prime $q > p$ such that q is even." The original statement is true.

7. Consider the claim: "$\forall a, b, c \in \mathbb{Z}^+,\ (a \mid bc \implies (a \mid b \vee a \mid c))$." (a) State the precise negation of this claim. (b) Find a counterexample with the minimal value of a. *(after TMUA 2021 Paper 2 Q4)*

Ans: (a) $\exists a, b, c \in \mathbb{Z}^+ : a \mid bc \wedge a \nmid b \wedge a \nmid c$. (b) $a = 4, b = 2, c = 2$.

8. Given that the compound implication $((P \vee Q) \implies R)$ is false, how many of the 8 possible truth assignments for (P, Q, R) are consistent with this fact?

Ans: 3.

9. Negate the statement: "$\forall x \in \mathbb{R},\ (x \in \mathbb{Q} \vee (x \notin \mathbb{Q} \wedge x > 0))$." Give a numerical witness proving the negation is true.

Ans: Negation: "$\exists x \in \mathbb{R} : x \notin \mathbb{Q} \wedge x \leq 0$." Witness: $-\sqrt{2}$ (or any non-positive irrational).

10. Negate: "For every prime p, $p^2 + 2$ is prime." What is the smallest prime counterexample to this statement? *(after TMUA 2022 Paper 2 Q3)*

Ans: Negation: "There exists a prime p such that $p^2 + 2$ is composite." Smallest counterexample: $p = 2$.

Section C Substitution & Structure

1. For real x and positive integer n, consider: I. For all x and for all n, $x^2 < n$. II. For all x, there exists n such that $x^2 < n$. III. There exists x such that for all n, $x^2 < n$. Which of I, II, III are true? *(after TMUA 2022 Paper 2 Q10)*

Ans: II and III only.

2. Determine the complete set of real values of k for which the following is true: "For all real x, if $x < k$, then $x^2 > k$." *(after TMUA 2022 Paper 2 Q9)*

Ans: C) $k \leq 0$.

3. Which is the correct negation of "For all primes p, p is odd or $p = 2$"?

Ans: B) There exists a prime p such that p is not odd and $p \neq 2$.

4. Negate: "There exists a positive integer n such that $n^2 + n + 1$ is even." Is the negation true or false?

Ans: Negation: "For all positive integers n, $n^2 + n + 1$ is odd." The negation is true.

5. Negate: "There exists a positive integer n such that for every positive integer $k \leq 10$, k divides n." Is the original statement true or false?

Ans: Negation: "For every positive integer n, there exists a positive integer $k \leq 10$ such that k does not divide n." The original statement is true.

6. Consider the statement about a real number x: "There exists a real number y such that $xy-x+y$ is positive." For how many real values of x is this true? *(after TMUA 2022 Paper 2 Q13)*

Ans: E) all values of x.

7. Negate: "There exists a real number x such that for every real number y, $xy = y$." Is the original statement true or false?

Ans: Negation: "For every real number x, there exists a real number y such that $xy \neq y$." The original statement is true.

8. Negate: "For every positive integer n, there exists a positive integer $m < n$ that is prime." Consider the cases $n = 1$ and $n = 2$ carefully. Is the original statement true or false?

Ans: Negation: "There exists a positive integer n such that for all positive integers $m < n$, m is not prime." The original statement is false.

Section D Challenge

1. Consider the statement: "For every finite sequence of positive integers, there exists a positive integer m such that every term of the sequence is at most m." Which of the following is the correct negation?

Ans: A) There exists a finite sequence of positive integers such that, for every positive integer m, some term exceeds m.

2. Determine, with brief justification, whether the following is true or false: "For every positive real number ε, there exists a positive integer N such that for every integer $n \geq N$, $\dfrac{1}{n} < \varepsilon$."

Ans: True.

3. A selection S of n terms is taken from the arithmetic sequence $2, 5, 8, 11, \ldots, 92$. What is the smallest value of n for which S must contain two distinct terms summing to 94? *(after TMUA 2022 Paper 2 Q8)*

Ans: C) 17.

4. Statement $(*)$: "For every finite set of primes $\{p_1, \ldots, p_k\}$, there exists a prime not in the set." (a) Negate $(*)$. (b) Is $(*)$ true or false? Justify briefly.

Ans: (a) "There exists a finite set of primes $\{p_1, \ldots, p_k\}$ such that every prime is in the set." (b) True.

5. Let $f : \mathbb{R} \to \mathbb{R}$. Consider: I. "For every x, there exists y such that $f(y) = x$." II. "There exists y such that for every x, $f(y) = x$." (a) Are I and II logically equivalent? (b) Give a concrete function f for which I is true but II is false.

Ans: (a) No. (b) $f(x) = x$ (or any surjective non-constant function).